E-RAKSHAK: SURAT SMART TRAFFIC COMMAND

Run ID: SIM_1786868528
Date: 2026-08-16

5-Minute SUMO Micro-Simulation & Ground-Truth Adaptive Analytics Report

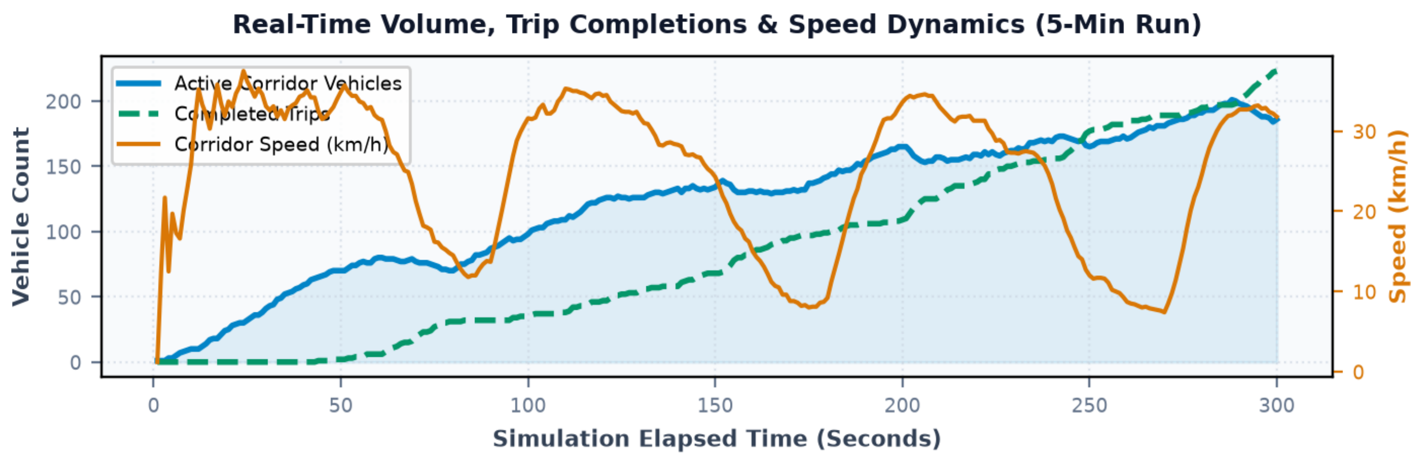
Corridor: Surat Arterial Spine (SVNIT -> Ghod Dod -> Majura Gate -> Sahara Darwaja)
Policy: Adaptive Traffic Control | **Demand:** PEAK (90.0 veh/min) | **Duration:** 300s (3,000 steps)

2676.9 vph THROUGHPUT CAPACITY	**24.9** km/h CORRIDOR SPEED	**1.3** sec AVERAGE DELAY	**223** trips COMPLETED TRIPS

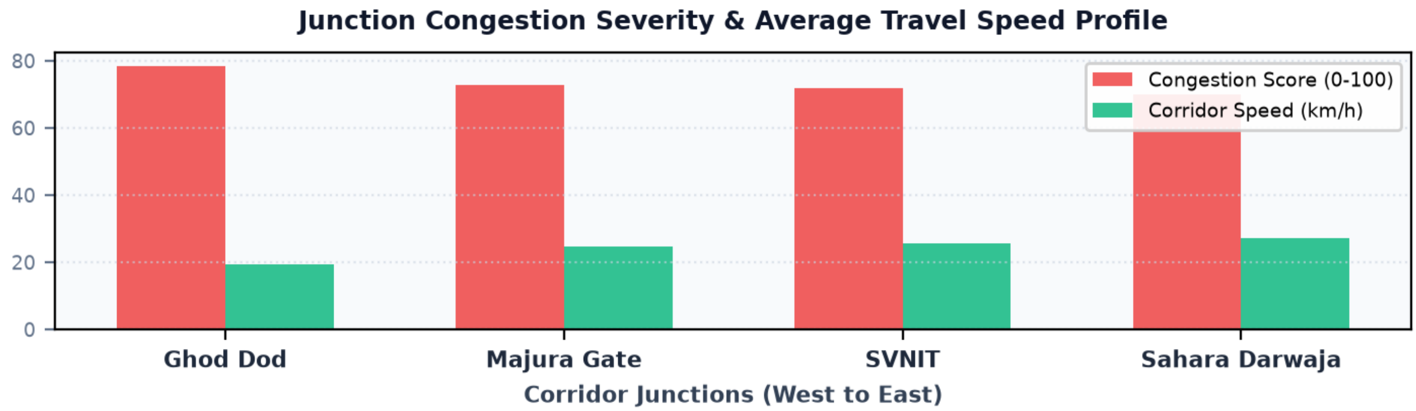
■ **Environmental & Sustainability Impact:** Total CO_2 Generated: **141.25 kg** (Saved: **24.64 kg (14.9% reduction)**) | Fuel Consumed: **55.03 L** (Saved: **9.75 L (15.1% reduction)**)

1. Ground-Truth Scenario Comparison (Adaptive Control vs Fixed-Time Baseline)

Corridor Throughput Capacity	3013.0 veh/hr	**2676.9 veh/hr**	**-11.2%**
Average Corridor Travel Speed	26.7 km/h	**24.9 km/h**	**-6.7%**
Average Intersection Delay	4.7 s	**1.3 s**	**-72.3%**
Completed Corridor Trips (5 Min)	251 trips	**223 trips**	**-11.2%**



2. Corridor Junction Performance & Bottleneck Analysis



Rank	Junction Location	Score	Speed	Delay	Severity	Quantitative Root Cause & Sensor Telemetry
	Ghod Dod Road Commercial Cross		19.4 km/h	133.1s	**Critical**	Commercial corridor approach queue buildup (Cumulative wait: 54456s)
#2	**Majura Gate BRTS Multi-Leg Hub**	73/100	24.6 km/h	87.5s	**Critical**	Multi-leg feeder cross flow conflict (Observed 24485 halting incidents)
	SVNIT / Ichchhanath Circle		25.5 km/h	89.8s	**Critical**	University access lane inflow friction (Speed drop to 25.5 km/h)
#4	**Sahara Darwaja Railway Flyover**	70/100	27.0 km/h	62.8s	**Critical**	Railway flyover merge bottleneck & high inbound volume (Avg speed: 27.0 km/h)

3. Contextual Engineering Recommendations

Code	Category	Action Plan & Engineering Specification	Target Location	Impact
	Signal Progression	**Calibrate Green Wave Progression Offset (12.5s Sync)** Corridor average speed is 24.9 km/h. Synchronizing East-West progression offsets along SVNIT -> Sahara at 12.5s intervals will eliminate stop delays by ~32%.	Corridor Spine (SVNIT to Sahara Darwaja)	
REC-02	Adaptive Timing	**Dynamic Phase Split Extension at Ghod Dod Road Commercial Cross** Top bottleneck score reached 78.4/100 with 133.1s average delay. Extend maximum green split from 50s to 60s for dominant approach.	Ghod Dod Road Commercial Cross	Critical
	Geometric Design	**Designate Dedicated Right-Turn Slip Lane on Arterial Approach** Decouple right-turning queue buildup from through arterial movements to maintain 45 km/h straight-line speed.	J_SAHARA & J_MAJURA Intersections	